INCH-POUND

MIL-DTL-46027N (MR)
29 February 2024
SUPERSEDING
MIL-DTL-46027M (MR)
w/AMENDMENT 1
13 September 2022

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM ALLOY, WELDABLE 5083, 5456, and 5059

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers AA5083, AA5456, & AA5059 wrought aluminum alloy armor plate for both welded and unwelded applications in nominal thicknesses from 0.250 to 3.000 inch, inclusive with additional greater thicknesses for AA5083 from 3.001 to 10.500 (see 6.2).

1.2 Weldability. All three classes of material covered by this specification have demonstrated to be weldable to itself and other weldable alloys (see 6.3).

1.3 Classification. The wrought aluminum armor should be of the following classes as specified (see 6.2).

1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 5083 aluminum alloy. The applicable gauge range for class I is 0.250-10.500 inches.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 5456 aluminum alloy. The applicable gauge range for class II is 0.250-3.000 inches.

1.3.3 Class III. Class III is wrought aluminum armor that conforms to the Aluminum Association designation for the 5059 aluminum alloy. The applicable gauge range for class III is 0.250-3.000 inches.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org/).

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at https://ansi.org/).

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ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling and Sample Preparation of Aluminum and Aluminum Alloys for Determination of Chemical Composition by Spark Atomic Emission Spectrometry
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Spark Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required, and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for classes I, II, and III alloys defined by the Aluminum Association Teal Sheets, “International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys” are listed within table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

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TABLE I. Chemical composition, percent. 1/

ELEMENTS	SYMBOL	CLASS I 5083 ALLOY 2/	CLASS II 5456 ALLOY 2/	CLASS III 5059 ALLOY 2/
Silicon	Si	0.40	0.25	0.45
Iron	Fe	0.40	0.40	0.50
Copper	Cu	0.10	0.10	0.25
Manganese	Mn	0.40 - 1.0	0.50 - 1.0	0.6 - 1.2
Magnesium	Mg	4.0 - 4.9	4.7 - 5.5	5.0 - 6.0
Chromium	Cr	0.05 - 0.25	0.05 - 0.20	0.25
Zinc	Zn	0.25	0.25	0.40 - 0.9
Titanium	Ti	0.15	0.20	0.20
Zirconium	Zr	---	---	0.05 - 0.25
Other, max. Each	---	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder

1/ Except for “Aluminum” and “others”, analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those “others” metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal (L) direction shall meet the minimum mechanical properties listed in table II. If mechanical property requirements differ from those contained in table II, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE II. Minimum mechanical properties. 1/,2/

Thickness, inches	Tensile Strength, ksi			Yield Strength, 0.2% Offset, ksi			Elongation percent		
	5083	5456	5059	5083	5456	5059	5083	5456	5059
0.250 to 0.499, incl.	45.0	54.0	53.0	35.0	47.0	39.0	8	6	8
0.500 to 2.000, incl.	45.0	50.0	52.0	37.0	45.0	39.0	8	6	7
2.001 to 3.000, incl.	44.0	50.0	47.0	35.0	44.0	39.0	9	8	8
3.001 to 10.500, incl.	43.0	TBD	TBD	31.0	TBD	TBD	11	TBD	TBD

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch.
2/ Values are taken in the Longitudinal Direction.

3.4 Weldability. When the vehicle requires weldments the fabricators (OEMs and kit manufacturers, etc.) are required to demonstrate weldability (which includes the Proofing Projectile Impact Testing (PPIT) formerly known as ballistic shock testing) for armor to armor and armor to non-armor applications in accordance with MIL-STD-3057, “Arc Welding of Armor Grade Aluminum”. Since the alloys referenced by this specification are not all listed in MIL-STD-3057, it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3. PPIT procedures and associated striking velocities for a variety of class I and class III plate thicknesses are listed in MIL-STD-3057. For any newly

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introduced armor material, configuration, or unproven thickness being used for armor applications, the aluminum fabricators shall qualify a weld procedure in accordance with MIL-STD-3057, and this requires PPIT testing.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. No ballistic limit shall be specified for plate thickness of 0.250 to 0.499 inch, inclusive. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.6 Stress corrosion resistance. When stress corrosion testing is specified in the contract or purchase order (see 6.2) and unless otherwise specified in the contract or purchase order (see 6.2), plate 0.750 inch and over in thickness shall be resistant to stress corrosion cracking, with a minimum of 5 of the 9 specimens showing no evidence of cracking at the end of 96 hours.

3.7 Thermal processing. After final strain hardening performed during plate fabrication, any thermal processing exceeding 200°F (93°C) excluding fabrication by welding shall necessitate recertification for conformance to 3.3, 3.5 and 3.6.

3.7.1 Storage of plates prior to application. After final strain hardening performed during plate fabrication, any extended outdoor storage (including container storage, e.g. CONEXs) exceeding 3 years shall necessitate recertification for conformance to 3.3, 3.5, and 3.6 by the entity owning the plates. MIL-DTL-46027 plates with unknown chain of custody shall also be recertified. For plates procured and qualified prior to the date of publication, this storage requirement while not mandatory is recommended. For optimal long term storage precautions that may obviate this storage and subsequent conformance requirement (see 6.5).

3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2/H35.2M, except for thickness.

3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in table III.

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TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)								
		OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
		THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)								
0.250	0.315		0.012	0.014	0.015	0.018	0.022	0.027	0.035	0.043
0.315	0.394		0.015	0.017	0.020	0.023	0.027	0.033	0.041	0.051
0.394	0.630		0.023	0.023	0.027	0.032	0.035	0.043	0.053	0.065
0.630	0.984		0.031	0.031	0.037	0.043	0.047	0.058	0.070	0.085
0.984	1.575		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.150		0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.150	3.937		0.100	0.100	0.115	0.125	0.130	0.160	---	---
3.937	6.299		0.130	0.130	0.145	0.165	---	---	---	---
6.300	8.000		0.160	0.160	0.160	0.165	---	---	---	---
8.001	10.500		0.190	0.190	0.190	0.190	---	---	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.9.1 Ballistic test plates. In addition to the markings in 3.9, each ballistic test plate shall be marked with the letters ‘PRE’ for First Article test plates and ‘ACC’ for Acceptance test plates. This marking shall be impression stamped on the horizontal edge face of the ballistic plate in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor, a properly completed Aluminum Armor Test Data Form (see figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use. Minor scratches, gouges, dents and similar damage are acceptable if the thickness is not reduced below the minimum material tolerance thickness.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR									
SPECIFICATION: MIL-DTL-46027			**REVISION: M**					**AMENDMENT: N/A**	
FIRING RECORD:		**DATE:**		**TEST CONTRACT NO:**					
	PLATE MANUFACTURER / PRODUCER			**PRIME CONTRACTOR**					
Name:									
Address:									
POC:									
Phone No:									
Fax No:									
TEST ITEM IDENTIFICATION:				Ordered Thickness:					
Lot No. _____ **Plate No.** _____		**Alloy:** _____ **Temper:** _____		**Class:** _____ **Type:** _____					
PURPOSE: ☐ Acceptance ☐ First Article ☐ Development ☐ Reduced Testing ☐ Audit Testing									
SAMPLE: ☐ Primary ☐ Retest (Firing Record No. of Failed Sample)									
CHEMISTRY ANALYSIS	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**		**Other - Total**	**Al**
									Rem.
MECHANICAL PROPERTIES:									
UTS (ksi):		0.2% YS (ksi):		Elongation (%):					
WELDABILITY:	Thickness	Required Velocity	Actual Velocity	Pass/Fail					
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES** ☐ **NO** ☐									
Test:	**Obl. (deg)**	**Actual Thickness (in)**	**Required V50 (fps)**	**Actual V50 (fps)**	**Pass/ Fail**				
Projectile:									
	NOTES:								
SPALL:	Spall Size:	Two Projectile Calibers:	Pass / Fail:						
Lot [☐ met] [☐ failed to meet] the ballistic requirements of specification MIL-DTL-46027									
Government Representative:	Date:	Supplier Representative:	Date:						

Figure 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all production and ballistic plates of the same alloy and from the same Direct Chill (DC) Cast, of the same source of molten metal representing the ingots poured, and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the DC ingot cast.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use in production.

4.3.1 First article inspection. First article inspection, except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two (2) plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2). For ordered plate thicknesses exceeding 4.000”, the ballistic test plates shall be symmetrically machined from both top and bottom plate rolling surfaces to attain 4.000” (i.e. centered around the T/2 plane) with a thickness tolerance of +/- 0.060".

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4.5.1.4 Stress corrosion tests. Nine (9) stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.5.

4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

4.5.2.2.1 Thermal stability. When verification of thermal stability is required, mechanical testing of thermally stressed specimens shall be conducted. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot and thermally prepared prior to testing as directed in 4.7.2.1. Only one tension test specimen shall be made from each sample. The samples shall meet the minimum acceptance requirements for mechanical properties in table 2 when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.4.

4.5.2.4 Stress corrosion tests. When required (see 6.2), one plate, 12 inches by 12 inches, shall be randomly selected from each lot for stress corrosion testing. The plate shall meet the requirements when tested as specified in 4.7.5.

4.6 Examination.

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4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.8) and workmanship (see 3.11) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.7 in accordance with the sampling procedures approved by the procuring activity (see 6.2).

4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of any dispute, analysis by method E3061shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal (L) direction. For material less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.2.1 Thermal stability specimens. When demonstration of thermal stability is required (see 4.5.2.2.1), mechanical properties of artificially aged specimens shall be determined (see 6.2(d)), specimens will follow the preparation of 4.7.2. The tensile specimens shall be artificially aged at 212°F (100°C) for 24 hours, removed and passively air cooled (without forced air) to 77°F (25°C), and then the tension tests shall be performed in accordance with ASTM B557.

4.7.3 Weldability. The thickness of the weldments specified in the contract or drawing shall be used to determine the proofing projectile and striking velocity in accordance with MIL-STD-3057.

4.7.4 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL (P) for that plate.

4.7.4.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to Rock Island Arsenal - Joint Manufacturing & Technology Center, Simulation & Test Range Division, Building 25 Prospect Court, Rock Island, IL 61299-5000 or to an approved Government facility for ballistic testing for first article or production acceptance. If Rock Island is unavailable or contraindicated in the contract or purchase order then the ballistic test plates shall be forwarded to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059.

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4.7.4.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees
0.500 - 0.749	Cal. .30 AP	30
0.750 - 1.000	Cal. .50 FSP	0
1.001 - 1.700 1/	20mm FSP	0
1.001 - 1.700 1/	Cal. .30 AP M2	0
1.701 - 2.000	Cal. .30 AP M2	0
2.001 - 10.500	Cal. .50 AP M2	0

1/ Two (2) types of projectiles for the same thickness range.

4.7.5 Capability requirements for resistance to stress corrosion. When stress corrosion testing is required (see 6.2), specimens shall be prepared in accordance with ASTM G38 and tested in accordance with ASTM G47 with nine specimens each prepared per sample per lot. The stress corrosion specimens shall be stressed to 30 ksi.

4.8 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection, except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense test thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.5. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.4 and 4.7.5.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive

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lots indicate that a uniform product meeting the testing requirements is being produced and providing that the manufacturer agrees in writing to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Alloys 5083, 5456 and 5059 will be considered for multiple applications that require weldability, ballistic and blast performance.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3).
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties (to include thermal stability mechanical properties) and ballistic requirements if different (see 3.3).
(e) When stress corrosion testing is required and if the test conditions are different (see 3.6, 4.5.2.4, and 4.7.5).
(f) Dimension and tolerance requirements, if different (see 3.8 and 3.8.1).
(g) If markings are different and when impression stamping is permissible (see 3.9).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.11.1).
(i) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(j) The orientation of the ballistic plate is different (see 4.5.1.3 and 4.5.2.3).
(k) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(l) If approval was requested and received for a different ballistic testing facility (see 4.7.4.1).
(m) Rejection and retest requirements, if different (see 4.8).
(n) If reduced testing is allowed (see 4.9).
(o) Packaging requirements (see 5.1).
(p) Striking velocities for those thicknesses not covered (see MIL-STD-3057).

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6.3 Weldability. The U.S. Army DEVCOM, Ground Vehicle Systems Center, DEVCOM ARL, and the Aberdeen Test Center have conducted testing and determined that class I and class III alloys can be welded and demonstrate good shock resistance when tested with the method described in in MIL-STD-3057, "Arc Welding of Armor Grade Aluminum". Qualification of new plate thicknesses and configurations under PPIT may be achieved in accordance with MIL-STD-3057, section 5.3.6.3.9: Proofing projectile and velocity. Additional tests have demonstrated that material covered by this specification is weldable to itself and other weldable 7000, 5000, and 6000 series alloys using a 5000 series filler metal. It is not weldable to 2000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated in accordance with MIL-STD-3057 prior to acceptance by the procuring activity.

6.4 Density. The density of the specified alloys listed in this specification are as follows:

AA5083 = 0.096 lb/in^3 (2.66 g/cc)
AA5456 = 0.096 lb/in^3 (2.66 g/cc)
AA5059 = 0.096 lb/in^3 (2.66 g/cc)

6.5 Long Term Storage. Al-Mg (5XXX) alloys are susceptible to degradation of mechanical properties and corrosion resistance after extended durations at elevated temperatures beyond 131°F (55°C) and additional preparations for long term storage are recommended. Due to surface heating produced by direct sunlight and/or cumulative trapped heat in poorly ventilated CONEX type shipping and storage containers, ventilated indoor warehouse storage is recommended for optimal storage of MIL-DTL-46027 qualified plates. Outdoor storage that provides overhead shading from direct sunlight is also acceptable. Horizontally stacked plates should be separated by nonmetallic spacers to facilitate easy access and manipulation by heavy equipment and to provide optimum airflow to dissipate heat and maintain lower temperatures. Any MIL-DTL-46027 qualified plates with full documentation of the recommended long term storage methods listed above are eligible to waive the recertification requirements in 3.7.1.

6.6 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

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6.7 Subject term (key word) listing.

Ballistic testing	M1114 HMMWV
Caliber .30 AP M2	Military vehicles
Caliber .50 AP M2	Stress corrosion
Caliber .50 FSP	20-mm FSP

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

ARMOR PLATE, ALUMINUM ALLOY, WELDABLE 5083, 5456, & 5059

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, weldable, 5083, 5456, and 5059 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of this document are available online at https://quicksearch.dla.mil/).

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

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A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-V.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the

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high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at their own expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at their own expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.460	1290	0.570	1452	0.680	1599
0.465	1298	0.575	1459	0.685	1605
0.470	1305	0.580	1466	0.690	1612
0.475	1313	0.585	1473	0.695	1618
0.480	1321	0.590	1480	0.700	1624
0.485	1328	0.595	1487	0.705	1630
0.490	1336	0.600	1494	0.710	1637
0.495	1343	0.605	1500	0.715	1643
0.500 1/	1351	0.610	1507	0.720	1649
0.505	1358	0.615	1514	0.725	1655
0.510	1366	0.620	1521	0.730	1661
0.515	1373	0.625	1527	0.735	1668
0.520	1381	0.630	1534	0.740	1674
0.525	1388	0.635	1541	0.745	1680
0.530	1395	0.640	1547	0.749 2/	1685
0.535	1402	0.645	1554	0.750	1686
0.540	1410	0.650	1560	0.755	1692
0.545	1417	0.655	1567	0.760	1699
0.550	1424	0.660	1573	0.765	1705
0.555	1431	0.665	1580	0.770	1711
0.560	1438	0.670	1586	0.775	1717
0.565	1445	0.675	1593	0.780	1723

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness (See table A-II).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.720	1681	0.835	2046	0.950	2485
0.725	1696	0.840	2064	0.955	2506
0.730	1711	0.845	2081	0.960	2527
0.735	1725	0.850	2099	0.965	2549
0.740	1740	0.855	2117	0.970	2570
0.745	1755	0.860	2135	0.975	2592
0.750 [1/]	1770	0.865	2153	0.980	2614
0.755	1785	0.870	2171	0.985	2636
0.760	1801	0.875	2190	0.990	2658
0.765	1816	0.880	2209	0.995	2680
0.770	1832	0.885	2227	1.000 [2/]	2703
0.775	1848	0.890	2246	1.005	2726
0.780	1863	0.895	2265	1.010	2749
0.785	1879	0.900	2284	1.015	2772
0.790	1895	0.905	2304	1.020	2795
0.795	1912	0.910	2323	1.025	2819
0.800	1928	0.915	2343	1.030	2842
0.805	1945	0.920	2363	1.035	2866
0.810	1961	0.925	2383	1.040	2890
0.815	1978	0.930	2403	1.045	2914
0.820	1995	0.935	2423	1.050	2939
0.825	2012	0.940	2444	1.055	2963
0.830	2029	0.945	2464	1.060	2988

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness (See table A-III and table A-IV).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.980	1389	1.150	1664	1.320	1991
0.985	1397	1.155	1673	1.325	2001
0.990	1404	1.160	1682	1.330	2012
0.995	1412	1.165	1691	1.335	2022
1.001[1/]	1420	1.170	1700	1.340	2033
1.005	1427	1.175	1709	1.345	2043
1.010	1435	1.180	1718	1.350	2054
1.015	1443	1.185	1727	1.355	2065
1.020	1450	1.190	1736	1.360	2076
1.025	1458	1.195	1745	1.365	2087
1.030	1466	1.200	1755	1.370	2098
1.035	1474	1.205	1764	1.375	2109
1.040	1481	1.210	1773	1.380	2120
1.045	1489	1.215	1783	1.385	2131
1.050	1497	1.220	1792	1.390	2142
1.055	1505	1.225	1801	1.395	2153
1.060	1513	1.230	1811	1.400	2165
1.065	1521	1.235	1821	1.405	2176
1.070	1529	1.240	1830	1.410	2187
1.075	1537	1.245	1840	1.415	2199
1.080	1546	1.250	1849	1.420	2210
1.085	1554	1.255	1859	1.425	2222
1.090	1562	1.260	1869	1.430	2234
1.095	1570	1.265	1879	1.435	2245
1.100	1579	1.270	1889	1.440	2257
1.105	1587	1.275	1899	1.445	2269
1.110	1596	1.280	1909	1.450	2281
1.115	1604	1.285	1919	1.455	2293
1.120	1613	1.290	1929	1.460	2305
1.125	1621	1.295	1939	1.465	2317
1.130	1630	1.300	1949	1.470	2329
1.135	1638	1.305	1959	1.475	2341
1.140	1647	1.310	1970	1.480	2353
1.145	1656	1.315	1980	1.485	2366

[1/] Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.490	2378	1.570	2585	1.650	2809
1.495	2390	1.575	2598	1.655	2823
1.500	2403	1.580	2612	1.660	2838
1.505	2415	1.585	2625	1.665	2853
1.510	2428	1.590	2639	1.670	2867
1.515	2441	1.595	2653	1.675	2882
1.520	2453	1.600	2666	1.680	2897
1.525	2466	1.605	2680	1.685	2912
1.530	2479	1.610	2694	1.690	2928
1.535	2492	1.615	2708	1.695	2943
1.540	2505	1.620	2722	1.700 [2/]	2958
1.545	2518	1.625	2737	1.705	2973
1.550	2531	1.630	2751	1.710	2989
1.555	2544	1.635	2765	1.715	3004
1.560	2558	1.640	2780	1.720	3019
1.565	2571	1.645	2794	1.725	3035

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.980	1792	1.160	2005	1.340	2198
0.985	1798	1.165	2011	1.345	2203
0.990	1804	1.170	2016	1.350	2208
0.995	1810	1.175	2022	1.355	2213
1.001 1/	1817	1.180	2027	1.360	2218
1.005	1823	1.185	2033	1.365	2223
1.010	1829	1.190	2038	1.370	2229
1.015	1836	1.195	2044	1.375	2234
1.020	1842	1.200	2050	1.380	2239
1.025	1848	1.205	2055	1.385	2244
1.030	1854	1.210	2060	1.390	2249
1.035	1860	1.215	2066	1.395	2254
1.040	1866	1.220	2071	1.400	2259
1.045	1872	1.225	2077	1.405	2264
1.050	1878	1.230	2082	1.410	2269
1.055	1884	1.235	2088	1.415	2273
1.060	1890	1.240	2093	1.420	2278
1.065	1896	1.245	2098	1.425	2283
1.070	1902	1.250	2104	1.430	2288
1.075	1907	1.255	2109	1.435	2293
1.080	1913	1.260	2114	1.440	2298
1.085	1919	1.265	2120	1.445	2303
1.090	1925	1.270	2125	1.450	2308
1.095	1931	1.275	2130	1.455	2313
1.100	1937	1.280	2136	1.460	2318
1.105	1942	1.285	2141	1.465	2322
1.110	1948	1.290	2146	1.470	2327
1.115	1954	1.295	2151	1.475	2332
1.120	1960	1.300	2157	1.480	2337
1.125	1965	1.305	2162	1.485	2342
1.130	1971	1.310	2167	1.490	2347
1.135	1977	1.315	2172	1.495	2352
1.140	1983	1.320	2177	1.500	2356
1.145	1988	1.325	2183	1.505	2361
1.150	1994	1.330	2188	1.510	2366
1.155	2000	1.335	2193	1.515	2371

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.520	2376	1.700	2541	1.880	2697
1.525	2380	1.705	2545	1.885	2701
1.530	2385	1.710	2550	1.890	2705
1.535	2390	1.715	2554	1.895	2709
1.540	2394	1.720	2559	1.900	2713
1.545	2399	1.725	2563	1.905	2717
1.550	2404	1.730	2568	1.910	2722
1.555	2409	1.735	2572	1.915	2726
1.560	2413	1.740	2576	1.920	2730
1.565	2418	1.745	2581	1.925	2734
1.570	2423	1.750	2585	1.930	2738
1.575	2427	1.755	2590	1.935	2742
1.580	2432	1.760	2594	1.940	2746
1.585	2437	1.765	2598	1.945	2751
1.590	2441	1.770	2603	1.950	2755
1.595	2446	1.775	2607	1.955	2759
1.600	2450	1.780	2611	1.960	2763
1.605	2455	1.785	2616	1.965	2767
1.610	2460	1.790	2620	1.970	2771
1.615	2464	1.795	2624	1.975	2775
1.620	2469	1.800	2629	1.980	2779
1.625	2473	1.805	2633	1.985	2783
1.630	2478	1.810	2637	1.990	2787
1.635	2482	1.815	2641	1.995	2791
1.640	2487	1.820	2646	2.000 [2/]	2796
1.645	2492	1.825	2650	2.005	2800
1.650	2496	1.830	2654	2.010	2804
1.655	2501	1.835	2658	2.015	2808
1.660	2505	1.840	2663	2.020	2812
1.665	2510	1.845	2667	2.025	2816
1.670	2514	1.850	2671	2.030	2820
1.675	2519	1.855	2675	2.035	2824
1.680	2523	1.860	2680	2.040	2828
1.685	2528	1.865	2684	2.045	2832
1.690	2532	1.870	2688	2.050	2836
1.695	2537	1.875	2692	2.055	2840

[2/] Specification requirements end for this ordered thickness (See table A-V).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.950	2041	2.110	2148	2.270	2250
1.955	2045	2.115	2151	2.275	2253
1.960	2048	2.120	2155	2.280	2256
1.965	2051	2.125	2158	2.285	2259
1.970	2055	2.130	2161	2.290	2263
1.975	2058	2.135	2164	2.295	2266
1.980	2062	2.140	2168	2.300	2269
1.985	2065	2.145	2171	2.305	2272
1.990	2068	2.150	2174	2.310	2275
1.995	2072	2.155	2177	2.315	2278
2.001 1/	2075	2.160	2180	2.320	2281
2.005	2079	2.165	2184	2.325	2284
2.010	2082	2.170	2187	2.330	2287
2.015	2085	2.175	2190	2.335	2290
2.020	2089	2.180	2193	2.340	2293
2.025	2092	2.185	2196	2.345	2296
2.030	2095	2.190	2200	2.350	2299
2.035	2099	2.195	2203	2.355	2303
2.040	2102	2.200	2206	2.360	2306
2.045	2105	2.205	2209	2.365	2309
2.050	2109	2.210	2212	2.370	2312
2.055	2112	2.215	2216	2.375	2315
2.060	2115	2.220	2219	2.380	2318
2.065	2119	2.225	2222	2.385	2321
2.070	2122	2.230	2225	2.390	2324
2.075	2125	2.235	2228	2.395	2327
2.080	2128	2.240	2231	2.400	2330
2.085	2132	2.245	2234	2.405	2333
2.090	2135	2.250	2238	2.410	2336
2.095	2138	2.255	2241	2.415	2339
2.100	2142	2.260	2244	2.420	2342
2.105	2145	2.265	2247	2.425	2345

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.430	2348	2.595	2445	2.760	2538
2.435	2351	2.600	2447	2.765	2541
2.440	2354	2.605	2450	2.770	2543
2.445	2357	2.610	2453	2.775	2546
2.450	2360	2.615	2456	2.780	2549
2.455	2363	2.620	2459	2.785	2552
2.460	2366	2.625	2462	2.790	2554
2.465	2369	2.630	2465	2.795	2557
2.470	2372	2.635	2468	2.800	2560
2.475	2375	2.640	2470	2.805	2563
2.480	2378	2.645	2473	2.810	2565
2.485	2380	2.650	2476	2.815	2568
2.490	2383	2.655	2479	2.820	2571
2.495	2386	2.660	2482	2.825	2574
2.500	2389	2.665	2485	2.830	2576
2.505	2392	2.670	2487	2.835	2579
2.510	2395	2.675	2490	2.840	2582
2.515	2398	2.680	2493	2.845	2585
2.520	2401	2.685	2496	2.850	2587
2.525	2404	2.690	2499	2.855	2590
2.530	2407	2.695	2501	2.860	2593
2.535	2410	2.700	2504	2.865	2595
2.540	2413	2.705	2507	2.870	2598
2.545	2416	2.710	2510	2.875	2601
2.550	2419	2.715	2513	2.880	2604
2.555	2421	2.720	2516	2.885	2606
2.560	2424	2.725	2518	2.890	2609
2.565	2427	2.730	2521	2.895	2612
2.570	2430	2.735	2524	2.900	2614
2.575	2433	2.740	2527	2.905	2617
2.580	2436	2.745	2529	2.910	2620
2.585	2439	2.750	2532	2.915	2622
2.590	2442	2.755	2535	2.920	2625

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.925	2628	3.090	2708	3.255	2781
2.930	2630	3.095	2711	3.260	2783
2.935	2633	3.100	2713	3.265	2786
2.940	2636	3.105	2715	3.270	2788
2.945	2639	3.110	2717	3.275	2790
2.950	2641	3.115	2720	3.280	2792
2.955	2644	3.120	2722	3.285	2794
2.960	2647	3.125	2724	3.290	2796
2.965	2649	3.130	2726	3.295	2799
2.970	2652	3.135	2728	3.300	2801
2.975	2654	3.140	2731	3.305	2803
2.980	2657	3.145	2733	3.310	2805
2.985	2660	3.150	2735	3.315	2807
2.990	2662	3.155	2737	3.320	2809
2.995	2665	3.160	2740	3.325	2811
3.000 [2/]	2668	3.165	2742	3.330	2814
3.005	2670	3.170	2744	3.335	2816
3.010	2672	3.175	2746	3.340	2818
3.015	2675	3.180	2748	3.345	2820
3.020	2677	3.185	2751	3.350	2822
3.025	2679	3.190	2753	3.355	2824
3.030	2681	3.195	2755	3.360	2827
3.035	2684	3.200	2757	3.365	2829
3.040	2686	3.205	2759	3.370	2831
3.045	2688	3.210	2762	3.375	2833
3.050	2690	3.215	2764	3.380	2835
3.055	2693	3.220	2766	3.385	2837
3.060	2695	3.225	2768	3.390	2839
3.065	2697	3.230	2770	3.395	2841
3.070	2699	3.235	2772	3.400	2844
3.075	2702	3.240	2775	3.405	2846
3.080	2704	3.245	2777	3.410	2848
3.085	2706	3.250	2779	3.415	2850

[2/] Class II and class III specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.420	2852	3.565	2913	3.710	2973
3.425	2854	3.570	2915	3.715	2975
3.430	2856	3.575	2917	3.720	2977
3.435	2858	3.580	2919	3.725	2979
3.440	2861	3.585	2921	3.730	2981
3.445	2863	3.590	2923	3.735	2983
3.450	2865	3.595	2925	3.740	2985
3.455	2867	3.600	2928	3.745	2987
3.460	2869	3.605	2930	3.750	2989
3.465	2871	3.610	2932	3.755	2991
3.470	2873	3.615	2934	3.760	2993
3.475	2875	3.620	2936	3.765	2995
3.480	2877	3.625	2938	3.770	2997
3.485	2880	3.630	2940	3.775	2999
3.490	2882	3.635	2942	3.780	3001
3.495	2884	3.640	2944	3.785	3003
3.500	2886	3.645	2946	3.790	3005
3.505	2888	3.650	2948	3.795	3007
3.510	2890	3.655	2950	3.800	3009
3.515	2892	3.660	2952	3.805	3011
3.520	2894	3.665	2954	3.810	3013
3.525	2896	3.670	2956	3.815	3015
3.530	2898	3.675	2958	3.820	3017
3.535	2901	3.680	2960	3.825	3019
3.540	2903	3.685	2962	3.830	3021
3.545	2905	3.690	2964	3.835	3023
3.550	2907	3.695	2967	3.840	3025
3.555	2909	3.700	2969	3.845	3027
3.560	2911	3.705	2971	3.850	3029

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.855	3031	3.970	3077	4.085	3122
3.860	3033	3.975	3079	4.090	3124
3.865	3035	3.980	3081	4.095	3125
3.870	3037	3.985	3083	4.100	3127
3.875	3039	3.990	3085	4.105	3129
3.880	3041	3.995	3087	4.110	3131
3.885	3043	4.000 3/,4/	3088	4.115	3133
3.890	3045	4.005	3090	4.120	3135
3.895	3047	4.010	3092	4.125	3137
3.900	3049	4.015	3094	4.130	3139
3.905	3051	4.020	3096	4.135	3141
3.910	3053	4.025	3098	4.140	3143
3.915	3055	4.030	3100	4.145	3145
3.920	3057	4.035	3102	4.150	3147
3.925	3059	4.040	3104	4.155	3149
3.930	3061	4.045	3106	4.160	3151
3.935	3063	4.050	3108	4.165	3152
3.940	3065	4.055	3110	4.170	3154
3.945	3067	4.060	3112	4.175	3156
3.950	3069	4.065	3114	4.180	3158
3.955	3071	4.070	3116	4.185	3160
3.960	3073	4.075	3118	4.190	3162
3.965	3075	4.080	3120	---	---

3/ Class I specification requirements end for this ordered thickness.
4/ For plates ordered in 4.001” to 6.000”, deduct 70fps from the required BL(P) of the final machined thickness; for plates ordered in 6.001” to 10.500”, deduct 100fps from the required BL(P) of the final machined thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project 9535-2023-002)

Review activities:
Army - AR, AT, AV, TE

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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